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22CB601

FINANCIAL MANAGEMENT

Department: COMPUTER SCIENCE & BUSINESS SYSTEMS

Batch/Year: 2023-27 / III

Created by:

Dr.K.Sudha/ASP/CSBS

Ms. M. Sindhu/AP/CSBS

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1. Course Objectives

The Course will enable learners to:

- Understand the fundamental concepts of financial management
- Understand valuation of securities.
- Analyze operating and financial leverages.
- Comprehend and apply the concepts of capital budgeting.
- Understand cash management.

2. Pre requisite

22CB302 - Fundamendals Economics

22CB501 – Fundamentals of Management

22CB502 – Business Strategy

3. SYLLABUS

22CB601 FINANCIAL MANAGEMENT

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UNIT I INTRODUCTION 9

Introduction: Introduction to Financial Management - Goals of the firm - Financial Environments. Time Value of Money: Simple and Compound Interest Rates, Amortization, Computing more that once a year, Annuity Factor.

UNIT II VALUATION OF SECURITIES 9

Valuation of Securities: Bond Valuation, Preferred Stock Valuation, Common Stock Valuation, Concept of Yield and YTM. Risk & Return: Defining Risk and Return, Using Probability Distributions to Measure Risk, Attitudes Toward Risk, Risk and Return in a Portfolio Context, Diversification, The Capital Asset Pricing Model (CAPM)

UNIT III OPERATING & FINANCIAL LEVERAGE 9

Operating & Financial Leverage: Operating Leverage, Financial Leverage, Total Leverage, Indifference Analysis in leverage study. Cost of Capital: Concept, Computation of Specific Cost of Capital for Equity - Preference – Debt, Weighted Average Cost of Capital – Factors affecting Cost of Capital 4L

UNIT IV CAPITAL BUDGETING 9

Capital Budgeting: The Capital Budgeting Concept & Process - An Overview, Generating Investment Project Proposals, Estimating Project, After Tax Incremental Operating Cash Flows, Capital Budgeting Techniques, Project Evaluation and Selection - Alternative Methods. Working Capital Management: Overview, Working Capital Issues, Financing Current Assets (Short Term and Long Term- Mix), Combining Liability Structures and Current Asset Decisions, Estimation of Working Capital.

UNIT V CASH MANAGEMENT 9

Cash Management: Motives for Holding cash, Speeding Up Cash Receipts, Slowing Down Cash Payouts, Electronic Commerce, Outsourcing, Cash Balances to maintain, Factoring. Accounts Receivable Management: Credit & Collection Policies, Analyzing the Credit Applicant, Credit References, Selecting optimum Credit period.

TOTAL: 45 PERIODS

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4. COURSE OUTCOMES

On successful completion of the course, students will be able to:

CO1: Understand the fundamental concepts of financial management
CO2: Apply valuation of securities and calculate the risk & return in portfolio management.
CO3: Analyse the cost structure of a company using operating and financial leverages.
CO4: Develop capital budgets and to estimate working capital.
CO5: Apply cash management in business.

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5. CO-PO/PSO MAPPING

CO	CO DESCRIPTION	Knowledge level
CO1	Understand the fundamental concepts of financial management	K2
CO2	Apply valuation of securities and calculate the risk & return in portfolio management.	K4
CO3	Analyse the cost structure of a company using operating and financial leverages.	K2
CO4	Develop capital budgets and to estimate working capital.	K2
CO5	Apply cash management in business.	K3

CO-PO-PSO Mapping

CO	K	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO8	PO 9	PO 10	PO 11	PO 12	PSO1	PSO 2	PSO 3	PSO 4
CO 1	K2	3	2	-	-	-	2	-	2	-	-	-	-	3	-	1	2
CO 2	K4	2	-	-	-	-	2	-	2	-	-	-	-	2	-	-	-
CO 3	K2	3	-	3	-	-	-	3	-	-	-	-	-	**2**	**2**	-	-
CO 4	K2	2	-	3	-	-	-	2	-	-	-	-	-	**3**	**2**	-	-
CO 5	K3	3	-	3	-	-	-	3	-	-	-	-	-	**2**	**2**	-	-

6. LECTURE PLAN

S.No	Topic	No. of Periods	Mode of Delivery	Date
1	Introduction	1	PPT	
2	Valuation of Securities: Bond Valuation,	1	PPT	
3	Preferred Stock Valuation, Common Stock Valuation,	1	PPT	
4	Concept of Yield and YTM.	1	PPT	
5	Risk & Return: Defining Risk and Return,	1	PPT	
6	Using Probability Distributions to Measure Risk,	1	PPT	
7	Attitudes Toward Risk,	1	PPT	
8	Risk and Return in a Portfolio Context, Diversification,	1	PPT	
9	The Capital Asset Pricing Model (CAPM)	1	PPT	

Webinar

TOPIC:

- Make the students to explore their views and knowledge about Valuation of Securities
- Make them to understand the Risk & Return
- Make the students to actively participate in the deliberations.

BENEFITS:

- This webinar improves the confidence of the students in deliberating technical talks
- It also motivates each and everyone to participate in Technical paper presentation, conferences etc.

8. LECTURE NOTES

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Unit II - Valuation of Securities

INTRODUCTION

The goal of investors and investment managers is to maximize their rate of return, or in other words, market value of their investments. Thus, investment management is an ongoing process that calls for continuous monitoring of information that may affect the value of securities or rate of return of such securities.

Therefore, in making valuation judgements about securities, the analyst applies consistently a process which will achieve a true picture of a company over a representative time span, an estimate of current normal earning power and dividend pay-out; estimate of future profitability; growth and reliability of such expectations and the translation of all these estimates into valuation of the company and its securities.

We assume that the appropriate discount rate is known. The question of how risk determines the appropriate discount rate is very important and we will discuss this issue. For now, our focus will be on what the relevant cash flows of financial assets are and how to value them, given an appropriate discount rate.

The objective of financial management is to maximise the value of the firm. Hence managers must know how stocks and bonds are valued. Knowing how to value securities (bonds and stocks, in the main) is as important for investors as it is for managers.

Current and prospective investors must understand how to value bonds and stocks. Such knowledge is helpful to them in deciding whether they should buy or hold or sell securities at the prices prevailing in the market. The basic discounted cash flow valuation model and its application to bonds and stocks is discussed in valuation of securities.

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VALUATION CONCEPTS

The term value is used in different senses.

Liquidation Value versus Going Concern Value

- The **liquidation value** is the amount that can be realised when an asset, or a group of assets representing a part or even the whole of a firm, is sold separately from the operating organisation to which it belongs.
- In contrast, the **going concern value** represents the amount that can be realised if the firm is sold as a continuing operating entity.
- In general, security valuation models assume a going concern, an operating business entity that generates cash flows to its security holders. When the going concern, assumption is not appropriate as in the case of an impending bankruptcy, liquidation value of assets is more relevant in determining the worth of the firm's financial securities.

Book Value versus Market Value

- The book value of an asset is the accounting value of the asset, which is simply the historical cost of the asset less accumulated depreciation or amortisation as the case may be. The book value of a firm's equity is equal to the book value of its assets minus the book value of its liabilities. Because book value reflects a historical accounting value it may diverge significantly from market value. However, under IFRS accounting, which is expected to be adopted in India, more assets are likely to be reported at "fair values." So, the traditional divergence between book value and market value may diminish.
- The market value of an asset is simply the market price at which the asset trades in the market place. Often the market value is greater than the book value.

Market Value versus Intrinsic Value

- As the nomenclature suggests, the market value of a security is the price at which the security trades in the financial market.
- The intrinsic value of a security is the present value of the cash flow stream expected from the security, discounted at a rate of return appropriate for the risk associated with the security. Put differently, intrinsic value is economic value. If the market is reasonably efficient, the market price of the security should hover around its intrinsic value. The focus of this chapter is on establishing a security's intrinsic value.

BOND VALUATION

- A bond represents a contract under which a borrower promises to pay interest and principal on specific dates to the holders of the bond.
- Bonds are issued by a variety of organisations. The principal issuers of bonds in India are the central government, state governments, public sector undertakings, private sector companies, and municipal bodies.
- Bonds issued by the central government are called Treasury bonds. These are bonds which have maturities ranging upto 20 years. These bonds generally pay interest semi-annually.
- Presently, Treasury bonds dominate the Indian bond market in terms of market capitalisation, liquidity, and turnover.
- State government bonds are issued by the state governments. These bonds have maturities that generally range from 3 to 20 years and pay interest semi-annually.
- Bonds issued by companies are classified into two types: PSU (public sector undertakings) bonds and private sector bonds.
- PSU bonds are bonds issued by companies in which the central or state governments have an equity stake in excess of 50 percent. Some of these bonds enjoy a tax-free status whereas others are taxable.
- Private sector bonds are bonds issued by private sector companies. Bonds issued by companies, PSU bonds as well as private sector bonds, generally have maturity ranging from 1 year to 15 years and pay interest semi-annually

Terminology

In order to understand the valuation of bonds, we need familiarity with certain bond related terms

Par Value

This is the value stated on the face of the bond. It represents the amount the firm borrows and promises to repay at the time of maturity. Usually the par or face value of bonds issued by business firms is Rs.100. Sometimes it is Rs.1,000.

Coupon Rate and Interest

A bond carries a specific interest rate which is called 'the coupon rate'. The interest payable to the bond holder is simply: par value of the bond x coupon rate. Most bonds pay interest semi-annually. For example, a government security which has a par value of Rs.1,000 and a coupon rate of 11 percent pays an interest of Rs.55 every six months.

Maturity Period

Typically bonds have a maturity period of 1-15 years; sometimes they have longer maturity. At the time of maturity the par (face) value plus perhaps a nominal premium is payable to the bondholder.

Valuation Model

The value of a bond - or any asset, real or financial - is equal to the present value of the cash flows expected from it. Hence determining the value of a bond requires:

- An estimate of expected cash flows
- An estimate of the required return

To simplify our analysis of bond valuation we will make the following assumptions:

- The coupon interest rate is fixed for the term of the bond.
- The coupon payments are made annually and the next coupon payment is receivable exactly a year from now.
- The bond will be redeemed at par on maturity.

Given these assumptions, the cash flow for a non-callable bond (a bond that cannot be prematurely retired) comprises of an annuity of a fixed coupon interest and the principal amount payable at maturity. Hence the value of the bond is:

$$P = \sum_{t=1}^{n} \frac{C}{(1+r)^t} + \frac{M}{(1+r)^n}$$

where P is the value (in rupees),

n is the number of years to maturity,

C is the annual coupon payment (in rupees),

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r is the periodic required return,

M is the maturity value,

t is the time when the payment is received.

Since the stream of coupon payments is an ordinary annuity, we can apply the formula for the present value of an ordinary annuity. Hence the bond value is given by the formula:

$$P = C \times \text{PVIFA}_{r,n} + M \times \text{PVIF}_{r,n}$$

To illustrate how to compute the price of a bond, consider a 10-year, 12% coupon bond with a par value of Rs. 1,000. Let us assume that the required yield on this bond is 13%. The

cash flows for this bond are as follows:

- 10 annual coupon payments of Rs.120
- Rs.1000 principal repayment 10 years from now

The value of the bond is :

$$P = 120 \times \text{PVIFA}_{13\%,10\text{yrs}} + 1{,}000 \times \text{PVIF}_{13\%,10\text{yrs}}$$

= 120 x5.426 + 1,000 x0.295
= 651.1 +295

= Rs.946.1

Bond Values with Semi-annual Interest

Most of the bonds pay interest semi-annually. To value such bonds, we have to work with a unit period of six months, and not one year. This means that the bond valuation equation has to be modified along the following lines:

- The annual interest payment, C, must be divided by two to obtain the semi-annual interest payment.
- The number of years to maturity must be multiplied by two to get the number of half-yearly periods.
- The discount rate has to be divided by two to get the discount rate applicable to half-yearly periods.

With the above modifications, the basic bond valuation becomes:

$$P = \sum_{t=1}^{2n} \frac{C/2}{(1+r/2)^t} + \frac{M}{(1+r/2)^{2n}}$$

$$= C/2\,(\text{PVIFA}_{r/2,2n}) + M(\text{PVIF}_{r/2,2n})$$

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where

P is the value of the bond,

C/2 is the semi-annual interest payment,

r/2 is the discount

rate applicable to a half-year period,

M is the maturity value,

2n is the maturity period expressed in terms of half-yearly periods.

consider an 8 year, 12 percent coupon bond with a par value of Rs.100 on which interest is payable semi-annually. The required return on this bond is 14 percent.

Applying above equation, the value of the bond is :

$$P = \sum_{t=1}^{16} \frac{6}{(1.07)^t} + \frac{100}{(1.07)^{16}}$$

$$= 6(\text{PVIFA}_{7\%,16}) + 100(\text{PVIF}_{7\%,16})$$

$$= \text{Rs.}6(9.447) + \text{Rs.}100(0.339) = \text{Rs.}90.6$$

Relationship between Coupon Rate, Required Yield, and Price

- A basic property of a bond is that its price varies inversely with yield. The reason is simple. As the required yield decreases, the present value of the cash flow increases; hence the price increases. Conversely, when the required yield increases, the present value of the cash flow decreases.
- The price-yield relationship may be illustrated with an example.
- Consider a bond carrying a coupon rate of 14 percent issued 3 years ago for Rs. 1000 (its par value) by Signal Corporation.
- The original maturity of the bond was 10 years, so its residual maturity now is 7 years. The interest rate has fallen in the last 3 years and investors now expect a return of 10 percent from this bond. The price of this bond now would be

$$P_0 = \sum_{t=1}^{7} \frac{140}{(1.10)^t} + \frac{1000}{(1.10)^7} = \text{Rs.}1194.7$$

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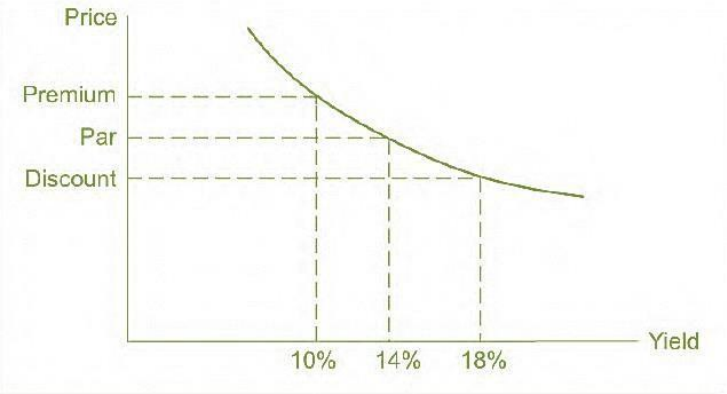
- The arithmetic of the bond price increase is clear. What is the logic behind it? The fact that

  the required return on such a bond has fallen to 10 percent means that if you had Rs.1,000 to invest, you can buy new bonds like Signal's except that these new bonds would pay Rs.100, rather than Rs.140, by way of interest. Naturally, as an investor you would prefer Rs. 140 to Rs. 100, so you would be willing to pay more than Rs. 1,000 for a Signal bond to enjoy its higher coupons.
- All investors would behave similarly and consequently the bond of Signal would be bid up in price to Rs. 1194.7. At that price it would provide a return of 10 percent, the rate the new bonds offer.
- Now let us look at what happens when the interest rate rises after the bond has been issued. Assume that because of a rise in interest rates, investors now expect a return of 18 percent from the Signal bond. The price of the bond would be:

$$P_0 = \sum_{t=1}^{7} \frac{140}{(1.18)^t} + \frac{1000}{(1.18)^7} = \text{Rs.}848.5$$

The graph of the price-yield relationship for the bond has a convex shape as shown in below Exhibit

Price-Yield Relationship



To sum up, the relationship between the coupon rate, the required yield, and the price is as follows:

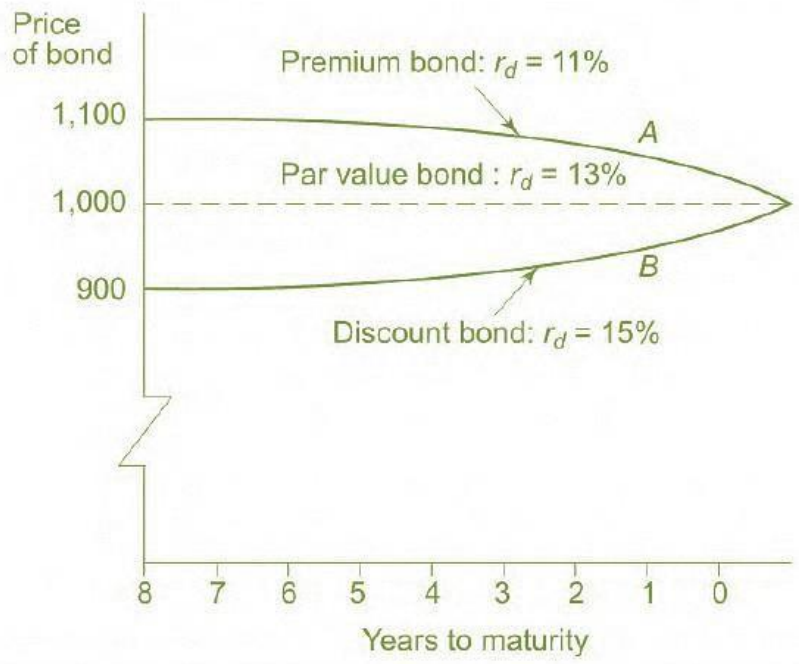
Coupon rate > Required yield ←----------> Price > Par (Premium bond)

Coupon rate = Required yield <----------> Price = Par

Coupon rate < Required yield ←----------> Price > Par (Discount bond)

Relationship between Bond Price and Time

- Since the price of a bond must typically be equal its par value at maturity (assuming that there is no risk of default), the bond price changes with time.
- For example, a bond that is redeemable for Rs. 1,000 (which is its par value) after 5 years when it matures, will have a price of Rs. 1,000 at maturity, no matter what the current price is. If its current price is, say, Rs.1,100, it is said to be a premium bond. If the required yield does change between now and the maturity date, the premium will decline over time as shown by curve A in Exhibit below.
- On the other hand, if the bond has a current price of say Rs.900, it is said to be a discount bond. The discount too will disappear over time as shown by curve B in Exhibit below.
- Only when the current price is equal to par value - in such a case the bond is said to be a par bond - there is no change in price as time passes, assuming that the required yield does not change between now and the maturity date.
- This is shown by the dashed line in Exhibit below



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BOND YIELDS

- In the previous section we learned how to determine the price of a bond and discussed how

  price and yield were related. We now discuss various yield measures.
- The commonly employed yield measures are: current yield, yield to maturity, and yield to call.

  Let us examine how these yield measures are calculated.

Current Yield

- The current yield relates the annual coupon interest to the market price. It is expressed as:

$$\text{Current yield} = \frac{\text{Annual interest}}{\text{Price}}$$

- For example, the current yield of a 10 year, 12 percent coupon bond with a par value of Rs.1000 and selling for Rs.950 is 12.63 percent.

$$\text{Current yield} = \frac{120}{950} = 0.1263 \text{ or } 12.63 \text{ percent}$$

- The current yield calculation reflects only the coupon interest rate. It does not consider the capital gain (or loss) that an investor will realise if the bond is purchased at a discount

  (or premium) and held till maturity. It also ignores the time value of money. Hence it is an

  incomplete and simplistic measure of yield.

Yield to Maturity

The yield to maturity (YTM) of a bond is the interest rate that makes the present value of the cash flows receivable from owning the bond equal to the price of the bond. Mathematically, it is the interest rate (r) which satisfies the equation:

$$P = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \ldots + \frac{C}{(1+r)^n} + \frac{M}{(1+r)^n}$$

where P is the price of the bond,

C is the annual interest (in rupees),

M is the maturity value (in rupees),

n is the number of years left to maturity.

The computation of YTM requires a trial and error procedure.

To illustrate this,

Consider a Rs. 1,000 par value bond, carrying a coupon rate of 9 percent, maturing after 8 years. The bond is currently selling for Rs.800. What is the YTM on this bond?

The YTM is the value of r in the following equation:

$$800 = \sum_{t=1}^{n} \frac{90}{(1+r)^t} + \frac{1000}{(1+r)^8}$$

$$= 90\,(\text{PVIFA}_{r,8\text{yrs}}) + 1{,}000\,(\text{PVIF}_{r,8\text{yrs}})$$

Let us begin with a discount rate of 12 percent. Putting a value of 12 percent for r we find that the right-hand side of the above expression is:

$$\text{Rs.}90\,(\text{PVIFA}_{12\%,8\text{yrs}}) + \text{Rs.}1{,}000\,(\text{PVIF}_{12\%,8\text{yrs}})$$
$$= \text{Rs.}90(4.968) + \text{Rs.}1{,}000(0.404) = \text{Rs.}851.0$$

Since this value is greater than Rs.800, we may have to try a higher value for r. Let us $r = 14$ percent. This makes the right-hand side equal to :

$$\text{Rs.}90\,(\text{PVIFA}_{14\%,8\text{yrs}}) + \text{Rs.}1{,}000\,(\text{PVIF}_{14\%,8\text{yrs}})$$
$$= \text{Rs.}90\,(4.639) + \text{Rs.}1{,}000\,(0.351) = \text{Rs.}768.1$$

Since this value is less than Rs.800, we try a lower value for r. Let us try $r = 13$ percent. T makes the right-hand side equal to :

$$\text{Rs.}90\,(\text{PVIFA}_{13\%,8\text{yrs}}) + \text{Rs.}1{,}000\,(\text{PVIF}_{13\%,8\text{yrs}})$$
$$= \text{Rs.}90\,(4.800) + \text{Rs.}1{,}000\,(0.376) = \text{Rs.}808$$

Thus r lies between 13 percent and 14 percent. Using a linear interpolation in the range percent to 14 percent, we find that r is equal to 13.2 percent.

$$13\% + (14\% - 13\%)\frac{808 - 800}{808 - 768.1} = 13.2\%$$

An Approximation

If you are not inclined to follow the trial-and-error approach described above, you can employ the following formula to find the approximate YTM on a bond :

$$\text{YTM} \simeq \frac{C + (M - P)/n}{0.4M + 0.6P}$$

Where,

YTM is the yield to maturity,

C is the annual interest payment,

M is the maturity value of the bond,

P is the present price of the bond,

n is the years to maturity.

To illustrate the use of this formula, let us consider the bond discussed above.

The approximate YTM of the bond works out to:

$$\text{YTM} = \frac{90 + (1000 - 800)/8}{0.4 \times 1000 + 0.6 \times 800} = 13.1\%$$

Thus, we find that this formula gives a value which is very close to the true value (13.2 percent). Hence it is very useful.

The YTM calculation considers the current coupon income as well as the capital gain or loss the investor will realise by holding the bond to maturity. In addition, it takes into account the timing of the cash flows.

Yield to Call

Some bonds carry a call feature that entitles the issuer to call (buy back) the bond prior to the stated maturity date in accordance with a call schedule (which specifies a call price for each call date).

For such bonds, it is a practice to calculate the yield to call (YTC) as well as the YTM.

The procedure for calculating the YTC is the same as for the YTM. Mathematically, the YTC is the value of r in the following equation:

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$$P = \sum_{t=1}^{n^*} \frac{C}{(1+r)^t} + \frac{M^*}{(1+r)^{n^*}}$$

where M^* is the call price (in rupees) and n^* is the number of years until the assumed date.

BOND MARKET

- Bonds are bought and sold in large quantities. The Indian bond market has grown rapidly since the mid - 1990s. With a daily turnover of about Rs.5,000 crore in mid-2003, it is one of the largest in Asia.
- The growth in the bond market has been stimulated by a host of reforms such as the increased functional autonomy of the RBI, improved institutional infrastructure, technology-related initiatives, and consolidation and creation of benchmark securities.
- Most trading in bonds takes place over the counter. This means that the transactions are privately negotiated and they don't take place through the process of matching of orders on an organised exchange.
- This is a characteristic of bond markets all over the world, not just in India. Because the bond market is largely over the counter, it lacks transparency. A financial market is transparent if you can easily observe its prices and volumes.
- The National Stock Exchange has a Wholesale Debt Market (WDM) segment. The WDM segment is a market for high value transactions in government securities,

  PSU bonds, commercial papers, and other debt instruments.

  The quotations of this segment mostly reflect over the counter transactions that are privately negotiated over the phone or computer and registered with the exchange for reporting purposes.
- The retail trade in corporate debt securities is done mostly on the capital market segment of the National Stock Exchange and the debt segment of the Bombay Stock Exchange.

Common stock valuation

- Common stock valuation is the process of determining the value of a share of stock in a company. The holder of one share in a company that has one million shares outstanding is actually the owner of one-millionth of the company; the value of that share should represent that percentage of the company's worth.
- Stock valuation is typically done using one of two basic approaches.
- The first is based on looking at fundamentals such as discounted cash flow and earnings analysis. This is similar to bond valuation, but the cash flows are more difficult to predict making stocks a riskier investment.
- The second is by studying market conditions — supply of and demand for the stock and general trends of the financial markets.
- In principle, the valuation of common stock is no different from the valuation of other types of securities, such as bonds and preferred stock. The basic procedure involves capitalizing (that is, discounting) the expected stream of cash flows to be received from holding the common stock. This is complicated by several factors, however.
- First, the expected cash flows from holding a common stock take two forms: the cash dividend payments made during the holding period and/or changes in the price of the stock (capital gains or losses) over the holding period. All the cash flows received by the common stockholder are derived from the firm's earnings and can be either paid to shareholders in the current period as cash dividends or reinvested in the firm to (it is hoped) provide higher future dividends and a higher stock price.

Preferred Stock Valuation

- Preferred stock generally pays regular, fixed dividends. Preference dividends are not increased when the profits of the firm rise, nor are they lowered or

  suspended unless the firm faces financial difficulties.
- If preference dividends are cut or suspended for some time, the firm is normally required to pay the arrears before paying equity dividends.
- Preference stock may be perpetual or redeemable. While the former has no maturity period, the latter is expected to be redeemed after its limited life.

  Preference stock in India is typically redeemable.
- If we assume that the preference stock pays fixed annual dividends during its life and the

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principal amount on maturity, its value is given as follows.

$$P_o = \sum_{t=1}^{n} \frac{D}{(1+r_p)^t} + \frac{M}{(1+r_p)^n}$$

where

P_0 is the current price of the preference stock,

D is the annual dividend,

n is the residual life of the preference stock,

r_p is the required rate of return on the preference stock,

M is the maturity value.

Since the stream of dividends is an ordinary annuity, we can apply the formula for the present value of an ordinary annuity. Hence the value of the preference stock is:

$$P_0 = D \times \text{PVIFA}r_{p,n} + M \times \text{PVIF}r_{p,n}$$

To illustrate how to compute the value of a preference stock, consider an 8 year, 10 percent preference stock with a par value of Rs. 1000. The required return on this preference stock is 9 percent.

The value of the preference stock is

$$P = 100 \times \text{PVIFA}_{9\%,\,8\text{yrs}} + 1000 \times \text{PVIF}_{9\%,\,8\text{yrs.}}$$
$$= 100 \times 5.535 + 1000 \times 0.502 = \text{Rs. } 1055.5$$

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STOCK MARKET

- The stock market consists of a primary segment and a secondary segment. New securities are issued in the primary market and outstanding securities are traded in the secondary market.
- The process of issuing securities in the primary market. Here our focus is mainly on the secondary market.
- The secondary market in India comprises of about twenty-three stock exchanges recognised by the government under the Securities Contracts (Regulation) Act. Of course, the principal bourses are the National Stock Exchange and the Bombay Stock Exchange, accounting for virtually all the trading on the Indian stock market.

Veritable Transformation

- The secondary market in India has undergone a metamorphosis after 1994. The following changes have virtually transformed the character of the secondary market.

Screen-based Trading

Till 1994, trading on the stock market in India was based on the open outcry system. With the establishment of the National Stock Exchange in 1994, India entered the era of screen-based trading. Within a short span of time, screen-based trading has supplanted the open outcry system on all the stock exchanges in the country, thanks to SEBTs initiative and the inherent superiority of screen-based trading.

Electronic Delivery

Traditionally, trades in India required physical delivery. This led to high paperwork cost and created bad paper risk. To mitigate the costs and risks associated with physical delivery, security transactions are now settled mainly through electronic delivery facilitated by depositories. A depository is an institution which dematerialises physical certificates and effects transfer of ownership by electronic book entries.

Rolling Settlement

Till recently security transactions in India were settled on the basis of a weekly account period. From 2002 onwards, SEBI has gradually introduced the rolling settlement system under which each day constitutes an account period and its trades are settled after a few days. For example, under the T + 2 rolling system which is currently in vogue inIndia, the trades are settled after 2 days.

Thanks to screen-based trading, electronic delivery, and rolling settlement, the transaction costs in India's stock market have decreased dramatically.

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Principal Exchanges

The National Stock Exchange has in a short period of time emerged as the largest exchange; the Bombay Stock Exchange, traditionally the leading exchange, is now the second largest exchange in the country. Hence it is instructive to understand the distinctive features of these principal exchanges.

The National Stock Exchange

The distinctive features of the National Stock Exchange (NSE), as it functions currently, are as follows:

- The NSE is a ringless, national, computerised exchange.
- The NSE has two segments: the Capital Market segment and the Wholesale Debt Market segment. The Capital Market segment covers equities, convertible debentures, and retail trade in non-convertible debentures. The Wholesale Debt Market segment is a market for high value transactions in government securities, PSU bonds, commercial papers, and other debt instruments.
- The trading members in the Capital Market segment are connected to the central computer in Mumbai through a satellite link-up, using VSATs (Very Small Aperture Terminals). The trading members in the Wholesale Debt Market segment are linked through dedicated high speed lines to the central computer at Mumbai.
- The NSE has opted for an order-driven system.
- When a trade takes place, a trade confirmation slip is printed at the trading member's workstation. It gives details like quantity, price, code number of counterparty, and so on.
- The identity of a trading member is not revealed to others.
- All trades on NSE are guaranteed by the National Securities Clearing Corporation (NSCC).

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The Bombay Stock Exchange

Established in 1875, the Bombay Stock Exchange (BSE) is one of oldest organised exchanges in the world with a long, colourful, and chequered history. Its distinctive features are as follows:

- The BSE switched from the open outcry system to the screen-based system in 1995. It accelerated its computerisation programme in response to the threat from the NSE.
- Jobbers traditionally played an important role on the BSE. A jobber is a broker who trades on his own account and hence offers a two-way quote or a bid-ask quote. The bid price reflects the price at which the jobber is willing to buy and the ask price represents the price at which the jobber is willing to sell.
- Investors have to transact via a jobber/broker. The jobber/broker feeds his buy/sell quotes in his computer terminal, which is linked to the main server at the BSE. Since both jobbers and brokers feed their orders, the BSE initially adopted a 'quote-driven' system and an 'order-driven' system, but subsequently shifted to the latter system.

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Risk and Return

What Is Risk?

In finance and economics, risk is a term that's related to uncertainty about an event and its outcome, regardless of whether the event and outcome are positive or negative. A good example of this is the risk of making a financial investment. We are uncertain about the outcome of investing in a stock and may quantify our uncertainty of loss and/or gain via a probability distribution model.

However, some models of risk are quite subjective. This is because some of our assumptions about risk deal with a person's individual attitudes toward risk and their understanding of a specific situation.

Introduction

- The value of a firm is affected by two key factors: risk and return. Higher the risk, other things being equal, lower the value; higher the return, other things being equal, higher the value.
- While intuitively the meaning of risk and return is grasped by almost every person who reflects on his experiences, the financial manager needs an explicit and quantitative understanding of these concepts, and, more importantly, the nature of relationship between them.
- Risk is present in virtually every decision. When a production manager selects an equipment, or a marketing manager an advertising campaign, or a financial manager a portfolio of securities, all of them face uncertain cash flows.
- Risk refers to the dispersion of a probability distribution: How much do individual outcomes deviate from the expected value? A simple measure of dispersion is the range of possible outcomes, which is simply the difference between the highest and lowest outcomes. A more sophisticated measure of risk, employed commonly in finance, is standard deviation.
- Assessing risks and incorporating the same in the final decision is an integral part of financial analysis.
- The objective in decision making is not to eliminate or avoid risk—often it may be neither

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feasible nor desirable to do so - but to properly assess it and determine whether it is worth bearing.

- Once the risk characterising future cash flows is properly measured, an appropriate risk-adjusted discount rate should be applied to convert future cash flows into their present values.
- Financial assets are expected to generate cash flows and hence the riskiness of a financial asset is measured in terms of the riskiness of its cash flows.
- The riskiness of an asset may be measured on a stand-alone basis or in a portfolio context.
- An asset may be very risky if held by itself but may be much less risky when it is part of a large portfolio.
- In the context of a portfolio, the risk of an asset is divided into two parts : diversifiable risk and market risk.
- Diversifiable risk arises from company-specific factors and hence can be washed away through diversification.
- Market risk, on the other hand, stems from general market movements and hence cannot be diversified away. For a diversified investor what matters is the market risk and not the diversifiable risk.
- In general, investors are risk-averse. So, they want to be compensated for bearing market risk. In a well-ordered market there is a linear relationship between market risk and expected return.
- Risk and return may be defined in relation to a single investment or a portfolio of investments.

HISTORICAL RETURNS AND RISK

- Whenever you make an investment decision, it helps to estimate the distribution of returns expected in future. Before we work with prospective returns, let us understand how to compute historical returns.

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Computing Historical Returns

The total return on an investment for a given period is:

$$\text{Total return} = \frac{\text{Cash payment received during the period} + \text{Price change over the period}}{\text{Price of the investment at the beginning}}$$

All items are measured in rupees. The rupee cash payment received during the period may be positive or zero. The rupee price change over the period is simply the difference between the ending price and the beginning price. This can be positive (ending price exceeds the beginning price) or zero (ending price equals the beginning price) or negative (ending price is less than the beginning price).

In formal terms

$$R = \frac{C + (P_E - P_B)}{P_B}$$

where

R is the total return over the period,

C is the cash payment received during the period,

P_E is the ending price of the investment,

P_B is the beginning price.

To illustrate, consider the following information for an equity stock:

- Price at the beginning of the year : Rs.60.00
- Dividend paid at the end of the year: Rs. 2.40
- Price at the end of the year : Rs.69.00

The total return on this stock is calculated as follows:

$$\frac{2.40 + (69.00 - 60.00)}{60.00} = 0.19 \text{ or } 19 \text{ percent}$$

It is helpful to split the total return into two components, viz., current yield and capital gains/loss yield as follows:

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$$\underbrace{\frac{\text{Cash payment}}{\text{Beginning price}}}_{\text{Current return}} + \underbrace{\frac{\text{Ending price} - \text{beginning price}}{\text{Beginning price}}}_{\text{Capital return}}$$

The total return of 19 percent in our example may be broken down as follows:

$$\frac{2.40}{60.00} + \frac{69.00 - 60.00}{60.00} = \underbrace{4 \text{ percent}}_{\text{Current return}} + \underbrace{15 \text{ percent}}_{\text{Capital return}}$$

Thus, the total return concept is all-inclusive (as it includes the current yield as well as the price change) and measures the total return per rupee of original investment. Hence it can be used for comparing investment returns over a specified period.

Using Probability Distributions to Measure Risk

EXPECTED RETURN AND RISK OF A SINGLE ASSET

- When you invest in a stock you know that the return from it can take various possible values. For example, it may be -5 per cent, or 15 per cent, or 35 per cent. Further, the likelihood of these possible returns can vary. Hence, you should think in terms of a probability distribution.
- The probability of an event represents the likelihood of its occurrence. Suppose you say that there is a 4 to 1 chance that the market price of a stock A will rise during the next fortnight.
- This implies that there is an 80 per cent chance that the price of stock A will increase and a 20 per cent chance that it will not increase during the next fortnight.
- Your judgment can be represented in the form of a probability distribution as follows:

Outcome	*Probability*
Stock price will rise	0.80
Stock price will not rise	0.20

- Another example may be given to illustrate the notion of probability distribution. Consider two equity stocks,
- Bharat Foods stock and Oriental Shipping stock.
- Bharat Foods stock may provide a return of 16 per cent, 11 per cent, or 6 per cent with certain probabilities associated with them, based on the state of the economy.

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- The second stock. Oriental Shipping stock, being more volatile, may earn a return of 40 per cent, 10 per cent, or -20 per cent with the same probabilities, based on the state of the economy.
- The probability distributions of the returns on these two stocks are shown in below Exhibit

		Rate of Return (%)	
State of the Economy	*Probability of Occurrence*	*Bharat Foods*	*Oriental Shipping*
Boom	0.30	16	40
Normal	0.50	11	10
Recession	0.20	6	–20

- Based on the probability distribution of the rate of return, you can compute two key parameters, the expected rate of return and the standard deviation of rate of return.

Expected Rate of Return

The expected rate of return is the weighted average of all possible returns multiplied by their respective probabilities. In symbols,

$$E(R) = \sum_{i=1}^{n} p_i R_i$$

Where,

E(R) is the expected return,

R_i, is the return for the ith possible outcome,

p_i, is the probability associated with Rjf

n is the number of possible outcomes.

From the above Eqn, it is clear that E(R) is the weighted average of possible outcomes – each outcome is weighted by the probability associated with it.

- The expected rate of return on Bharat Foods stock is:

  $E(R_b) = (0.30)(16\%) + (0.50)(11\%) + (0.20)(6\%) = 11.5\%$

  Similarly, the expected rate of return on Oriental Shipping stock is:

  $E(R_0) = (0.30)(40\%) + (0.50)(10\%) + (0.20)(-20\%) = 13.0\%$

Standard Deviation of Return

Risk refers to the dispersion of a variable. It is commonly measured by the variance or the standard deviation.

The variance of a probability distribution is the sum of the squares of the deviations of actual returns from the expected return, weighted by the associated probabilities.

In symbols,

$$\sigma^2 = \Sigma p_i(R_i - E(R))^2$$

where σ^2 is the variance, R_i is the return for the ith possible outcome, p_i is the probability associated with the ith possible outcome, and $E(R)$ is the expected return.

As an illustration, the standard deviation of returns on Bharat Foods stock and Oriental Shipping stock are calculated in Exhibit below.

Illustration of the Calculation of Standard Deviation

Bharat Foods Stock						
State of the Economy	p_i	R_i	p_iR_i	R_i-$E(R)$	$(R_i$-$E(R))^2$	$p_i(R_i-E(R))^2$
1. Boom	0.30	16	4.8	4.5	20.25	6.075
2. Normal	0.50	11	5.5	–0.5	0.25	0.125
3. Recession	0.20	6	1.2	–5.5	30.25	6.050
			$\Sigma p_iR_i = 11.5$			$\Sigma p_i(R_i - E(R))^2 = 12.25$
			$\sigma = [\Sigma p_i (R_i - E(R))^2]^{1/2} = (12.25)^{1/2} = 3.5\%$			
Oriental Shipping Stock						
1. Boom	0.30	40	12.0	27.0	729.0	218.7
2. Normal	0.50	10	5.0	–3.0	0.25	4.5
3. Recession	0.20	–20	–4.0	–33.0	1089.00	217.8
			$\Sigma p_iR_i = 13.0$			$\Sigma p_i (R_i - E(R))^2 = 441.0$
			$\sigma = [\Sigma p_i(R_i - E(R))^2]^{1/2} = (441.0)^{1/2} = 21.0\%$			

Features of Standard Deviation

Looking at the calculation of standard deviation, we find that it has the following features:

- The differences between the various possible values and the expected value are squared. This means that values which are far away from the expected value have a much more effect on standard deviation than values which are close to the expected value.
- The squared differences are multiplied by the probabilities associated with the respective values. This means that the smaller the probability that a

particular value will occur, the lesser its effect on standard deviation.

■ The standard deviation is obtained as the square root of the sum of squared differences (multiplied by their probabilities). This means that the standard deviation and expected value are measured in the same units and hence the two can be directly compared.

Rationale for Standard Deviation

Why is standard deviation employed commonly in finance as a measure of risk? The principal reasons for using standard deviation seem to be :

■ If a variable is normally distributed, its mean and standard deviation contain all the information about its probability distribution.

■ If the utility of money is represented by a quadratic function (a function commonly suggested to represent diminishing utility of wealth), then the expected utility is a function of mean and standard deviation.

■ Standard deviation is analytically more easily tractable.

Attitudes Toward Risk

Individuals take investment decisions based on risk, return and time horizon of the investment. Personality types of individuals affect their attitude towards financial risk and in turn their investment intentions.

What is attitude to risk?

Choosing the right investments means balancing how much you are prepared to lose with how much you hope to earn. This means understanding risk and your attitude to risk – how much risk you are prepared to take with your money and assets in a particular timeframe. This can be affected by many factors, including your personal situation, age, goals and the current economic climate.

Different people have different attitudes toward the risk-return trade-off.

Risk averse

People are **Risk averse** when they shy away from risks and prefer to have as much security and certainty as is reasonably affordable in order to lower their discomfort level. They would be willing to pay extra to have the security of knowing that unpleasant risks would be removed from their lives.

Economists and risk management professionals consider most people to be risk averse. So, why do people invest in the stock market where they confront the possibility of losing everything? Perhaps they are also seeking the highest value possible for their pensions and savings and believe that losses may not be pervasive

- Usually, the higher the risk you are willing to take, the higher the potential returns could be – but equally, the higher the risk of losing your money too.
- Low risk investments are likely to offer lower rewards, but have less risk attached too. You have to choose the options you are comfortable with.
- Your attitude to risk can of course change over time. You might become much less risk averse if you came into a big sum of money, for example, or much more risk averse if you start to have a family and need to think about securing your children's future too.

Risk seeker

- A **Risk seeker**, on the other hand, is not simply the person who hopes to maximize the value of retirement investments by investing the stock market. Much like a gambler, a risk seeker is someone who will enter into an endeavor (such as blackjack card games or slot machine gambling) as long as a positive long run return on the money is possible, however unlikely.

Risk neutral

- Finally, an entity is said to be **Risk neutral** when its risk preference lies in between these two extremes. Risk neutral individuals will not pay extra to have the risk transferred to someone else, nor will they pay to engage in a risky endeavor. To them, money is money. They don't pay for insurance, nor will they gamble. Economists consider most widely held or publicly traded corporations as making decisions in a risk-neutral manner since their shareholders have the ability to diversify away risk—to take actions that seemingly are not related or have opposite effects, or to invest in many possible unrelated products or entities such that the impact of any one event decreases the overall risk.

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Risk Aversion and Required Returns

- You are lucky to be invited by the host of a television game show. After the usual introduction, the host shows two boxes to you. He tells you that one box contains Rs. 5,000 and the other box is empty. He does not tell you which one is which. The host asks you to open any one of the two boxes and keep whatever you find in it.

- You are not sure which box you should open. Sensing your vacillation, he says he will offer you a certain Rs.3,000 if you forfeit the option to open a box. You don't accept his offer. He raises his offer to Rs.3,500. Now you feel indifferent between a certain return of Rs.3,500 and a risky (uncertain) expected return of Rs.5,000. This means that a certain amount of Rs.3,500 provides you with the same satisfaction as a risky expected value of Rs.5,000. Thus your certainty equivalent (Rs.3,500) is less than the risky expected value (Rs.5,000).
- Empirical evidence suggests that most individuals, if placed in a similar situation, would have a certainty equivalent which is less than the risky expected value.
- The relationship of a person's certainty equivalent to the expected monetary value of a risky investment defines his attitude toward risk. If the certainty equivalent is less than the expected value, the person is risk-averse; if the certainty equivalent is equal to the expected value, the person is risk-neutral; finally, if the certainty equivalent is more than the expected value, the person is risk-loving.
- In general, investors are risk-averse. This means that risky investments must offer higher expected returns than less risky investments to induce people to invest in them. Remember, however, that we are talking about expected returns; the actual return on a risky investment may well turn out to be less than the actual return on a less risky investment.
- Put differently, risk and return go hand in hand. This indeed is a well-established empirical fact, particularly over long periods of time.

Risk and Return in a Portfolio Context

- Most investors invest in a portfolio of assets, as they do not want to put all their eggs in one basket. Hence, what really matters to them is not the risk and return of stocks in isolation, but the risk and return of the portfolio as a whole.
- A portfolio is composed of two or more securities. Each portfolio has risk-return characteristics of its own. A portfolio comprising securities that yield a maximum return for given level of risk or minimum risk for given level of return is termed as 'efficient portfolio'. In their Endeavour to strike a golden mean between risk and return the traditional portfolio managers diversified funds over securities of large number of companies of different industry groups.
- The risk of assets held in a portfolio is less risky than the same assets held in isolation.
- Most financial assets are actually held as part of portfolio - Example: banks, insurance companies and mutual funds are required by law to hold a diversified portfolio.
- The risk and return of an individual security should be analysed in terms of how that security affect risk and return of the portfolio in which is it held.

Portfolio

It is combination of two or more securities or assets

Portfolio Return

The expected return of a portfolio is a weighted average of the expected returns of the securities constituting the portfolio.

For Example,

We have two securities, i.e. Security A and Security B and we have also their corresponding expected returns and standard deviation.

	Security A	Security B
Expected Return	10%	12%
Standard Deviation	8%	5%

- If equal amounts of money are invested in the two securities, the expected return of the portfolio is (0.5)10% + (0.5)12% = 11%

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Portfolio Risk

From the previous example our portfolio risk will be:

(0.5)8% + (0.5)5% = 5.25%

Portfolio risk is not so simple. To understand portfolio risk we must understand covariance

Covariance

It is a statistical measure of the degree to which two variables move together.

There are three different kinds of covariance directions.

Positive Covariance

It implies that the two variables move in the same direction.

Negative Covariance

It implies that the two variables move in the opposite direction.

Zero Covariance

It implies that the two variables show no tendency to vary together in either a positive or negative direction.

- The riskiness of a portfolio depends much more on the paired security covariance than on the riskiness of the separate security covariance.
  High covariance lead to high portfolio risk and
  Low covariance lead to low portfolio risk.
- **Illustration**
  Most investors (individuals as well as institutions) hold portfolios of securities.
  Hence, a very pertinent question is: What happens when two or more securities are combined in a portfolio?
  To answer this question, let us consider an example.
  Suppose you have Rs. 1,00,000 to invest and you are considering two equity stocks, Alpha Company and Beta Industries. The returns on the equity stocks of Alpha and Beta for the preceding five years are shown in below Table and you expect the future returns on these stocks to be equal to their past returns. This means that on your investment of Rs.1,00,000 you expect to earn Rs. 12,000 per annum (because the mean return in the preceding five years was 12 per cent) on either of the securities individually or on a

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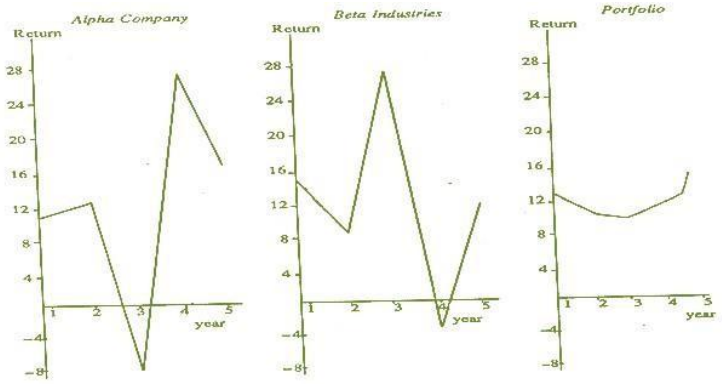
portfolio consisting of these securities.

Return and Risk:Individual Securities and Portfolio

Year	Alpha Company	Beta Industries	Portfolio:50%Alpha 50% Beta
1	11	15	13
2	13	9	11
3	-8	27	9.5
4	27	-3	12
5	17	12	14.5
Statistics@ Mean Return(k)	12%	12%	12%
Standard Deviation(s)	12.76%	10.81%	4.73%

What happens if you invest in a portfolio consisting of the equity stocks of Alpha Company and Beta Industries in equal proportions? While the expected return remains at 12 per cent the same as either company individually, the standard deviation is only 4.73 per cent, much less than the standard deviation of either stock individually.

As shown in below Fig, the variability in the portfolio rate of return is much less than the variability of individual security rates of return. Why? This happens because the rates of return on the two securities tend to move in opposite directions. In general, if the rates of return of individual securities are not perfectly positively correlated, diversification results in risk reduction.



@When historical return data, rather than the probability distribution of return, is used, the formulae used for calculating the mean return and the standard deviation

are as follows:

$$\text{Mean return} = \bar{k} = \sum_{i=1}^{n} k_i/n$$

$$\text{Standard deviation} = \sigma = \left[\sum (k_i - \bar{k})^2/n-1 \right]^{1/2}$$

Diversification

- Diversification is a technique of allocating portfolio resources or capital to a mix of different investments. The ultimate goal of diversification is to reduce the volatility of the portfolio by offsetting losses in one asset class with gains in another asset class. A phrase commonly associated with diversification: "Do not put all your eggs in one basket." Having "eggs" in multiple baskets mitigates risk, as if one basket breaks, not all eggs are lost.
- Diversification is a risk management strategy that mixes a wide variety of investments within a portfolio. A diversified portfolio contains a mix of distinct asset types and investment vehicles in an attempt at limiting exposure to any single asset or risk. The rationale behind this technique is that a portfolio constructed of different kinds of assets will, on average, yield higher long-term returns and lower the risk of any individual holding or security.
- Diversification is a technique that reduces risk by allocating investments across various financial instruments, industries, and other categories. It aims to maximize returns by investing in different areas that would each react differently to the same event.
- Most investment professionals agree that, although it does not guarantee against loss, diversification is the most important component of reaching long-range financial goals while minimizing risk.

Diversification and Unsystematic Risk

- Diversification is primarily used to eliminate or smooth unsystematic risk. Unsystematic risk is a firm-specific risk that affects only one company or a small group of companies. Therefore, when a portfolio is well-diversified, investments with a strong performance compensate for the negative results from poorly performing investments.
- However, diversification does not usually affect the inherent or systematic risk that applies to the financial markets as a whole.
- One way to think about the two basic types of risk is that one refers to the specific risks of an industry or individual firm, while the other refers to risk factors in the overall

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economy. Unsystematic risks can usually be controlled or mitigated, but systematic risk involves fundamental economic factors that are largely outside of any one individual company's power to control.

Portfolio Diversification

- Portfolio diversification concerns the inclusion of different investment vehicles with a variety of features. The strategy of diversification requires balancing various investments that have only a slight positive correlation with each other – or better yet, actual negative correlation. Low correlation usually means that the prices of the investments are not likely to move in the same direction.

- There is no consensus regarding the perfect amount of diversification. In theory, an investor may continue diversifying his/her portfolio virtually infinitely, as long as there are available investments in the market that are not correlated with other investments in the portfolio.

- An investor should consider diversifying his/her portfolio based on the following specifications:

Types of investments:
Include different asset classes, such as cash, stocks, bonds, ETFs, options, etc.
Risk levels:
Investments with dissimilar levels of risk allow the smoothing of gains and losses.
Industries:
Invest in companies from distinct industries. The stocks of companies operating in different industries tend to show a lower correlation with each other.

Foreign markets:
An investor should not invest only in domestic markets. There is a high probability that the financial products traded in foreign markets are less correlated with products traded in the domestic markets.

Nowadays, index and mutual funds, as well as ETFs, provide individual investors with a simple and inexpensive instrument for creating a diversified investment portfolio.

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Diversification and Portfolio Risk

Suppose you have Rs. 100,000 to invest and you want to invest it equally in two stocks, A and B. The returns on these stocks depend on the state of the economy. Your assessment suggests that the probability distributions of the returns on stocks A and B are as shown in below Exhibit. For the sake of simplicity, all the five states of the economy is assumed to be equiprobable. The last column of below Exhibit shows the return on a portfolio consisting of stocks A and B in equal proportions. The expected return and standard deviation of return on stocks A and B and the portfolio consisting of A and B in equal proportions are calculated in below Exhibit. (Since the portfolio in this case consists of only two securities, the proportion invested in security 2 is simply one minus the proportion invested in security 1)

Probability Distribution of Returns

State of the Economy	*Probability*	*Return on Stock A(%)*	*Return on Stock B(%)*	*Return on Portfolio(%)*
1	0.20	15	–5	5
2	0.20	–5	15	5
3	0.20	5	25	15
4	0.20	35	5	20
5	0.20	25	35	30

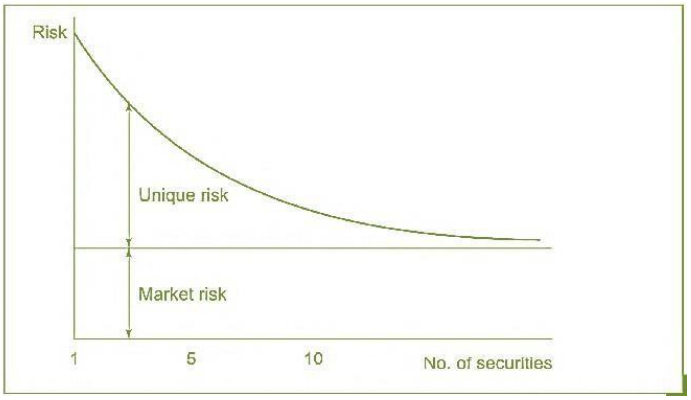
Exhibit below shows that if you invest only in stock A, the expected return is 15 percent and the standard deviation is 14.14 percent. Likewise, if you invest only in stock B, the expected return is 15 percent and the standard deviation is 14.14 percent. What happens if you invest in a portfolio consisting of stocks A and B in equal proportions? While the expected return remains at 15 percent, the same as that of either stock individually, the standard deviation of the portfolio return, 9.49 percent, is lower than that of either stock individually. Thus, in this case diversification reduces risk.

Expected Return and Standard Deviation

Expected Return				
Stock A	: 0.2(15%) + 0.2(–5%) + 0.2(5%) +0.2(35%) + 0.2(25%) = 15%			
Stock B	: 0.2(–5%) + 0.2(15%) + 0.2(25%) + 0.2(5%) + 0.2(35%) = 15%			
Portfolio of A and B	: 0.2(5%) + 0.2(5%) + 0.2(15%) + 0.2(20%) + 0.2(30%) = 15%			
Standard Deviation				
Stock A	:	σ_A^2	=	$0.2(15–15)^2 + 0.2(–5–15)^2 + 0.2(5–15)^2 + 0.2(35–15)^2 + 0.20\ (25–15)^2$
			=	200
		σ_A	=	$(200)^{1/2} = 14.14\%$
Stock B	:	σ_B^2	=	$0.2(–5–15)^2 + 0.2(15–15)^2 + 0.2(25–15)^2 + 0.2(5–15)^2 + 0.2\ (35–15)^2$
			=	200
		σ_B	=	$(200)^{1/2} = 14.14\%$
Portfolio	:	$\sigma^2_{(A+B)}$	=	$0.2(5–15)^2 + 0.2(5–15)^2 + 0.2(15–15)^2 + 0.2(20–15)^2 + 0.2(30–15)^2$
			=	90
		σ^2_{A+B}	=	$(90)^{1/2} = 9.49\%$

- In general, if returns on securities do not move in perfect lockstep, diversification reduces risk. In technical terms, diversification reduces risk if returns are not perfectly positively correlated.
- The relationship between diversification and risk is shown graphically in below Exhibit, Empirical studies suggest that the bulk of the benefit of
- diversification, in the form of risk reduction, is achieved by forming a portfolio of about 15 to 20 securities. Thereafter, the gain from diversification tends to be negligible.

Relationship between Diversification and Risk



Market Risk versus Unique Risk

Notice that the portfolio risk does not fall below a certain level, irrespective of how wide the diversification is. Why?

The answer lies in the following relationship which represents a basic insight of modern portfolio theory.

Total risk = Unique risk + Market risk

- The unique risk of a security represents that portion of its total risk which stems from firm specific factors like the development of a new product, a labour strike, or the emergence of a new competitor. Events of this nature primarily affect the specific firm and not all firms in general. Hence, the unique risk of a stock can be washed away by combining it with other stocks. In a diversified portfolio, unique risks of different stocks tend to cancel each other – a favourable development in one firm may offset an adverse happening in another and vice versa. Hence, unique risk is also referred to as diversifiable risk or unsystematic risk.

- The market risk of a stock represents that portion of its risk which is attributable to economy- wide factors like the growth rate of GNP, the level of government spending, money supply, interest rate structure, and inflation rate. Since these factors affect all firms to a greater or lesser degree, investors cannot avoid the risk arising from them, however diversified their portfolios may be. Hence, it is also referred to as systematic risk (as it affects all securities) or non-diversifiable risk.

The Capital Asset Pricing Model (CAPM)

- The Capital Asset Pricing Model (CAPM) describes the relationship between systematic risk and expected return for assets, particularly stocks. CAPM is widely used throughout finance for pricing risky securities and generating expected returns for assets given the risk of those assets and cost of capital.
- No matter how much you diversify your investments, some level of risk will always exist. So, investors naturally seek a rate of return that compensates for that risk. The capital asset pricing model (CAPM) helps to calculate investment risk and what return on investment an investor should expect.
- The CAPM, in essence, predicts the relationship between
- the risk of an asset and its expected return. This relationship is very useful in two important ways.
- First, it produces a benchmark for evaluating various investments. For example, when we are analysing a security, we are interested in knowing whether the expected return from it is in line with its fair return as per the CAPM.

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- Second, it helps us to make an informed guess about the return that can be expected from an asset that has not yet been traded in the market. For example, how should a firm price its initial public offering of stock?
- Although the empirical evidence on the CAPM is mixed, it is widely used because of the valuable insight it offers and its accuracy is deemed satisfactory for most practical applications.
- The capital asset pricing model provides a useful measure that helps investors determine what return they deserve on an investment, in exchange for putting their money at risk on it.
- The capital asset pricing model (CAPM) is an idealized portrayal of how financial markets price securities and thereby determine expected returns on capital investments. The model provides a methodology for quantifying risk and translating that risk into estimates of expected return on equity.
- A principal advantage of CAPM is the objective nature of the estimated costs of equity that the model can yield. CAPM cannot be used in isolation because it necessarily simplifies the world of financial markets. But financial managers can use it to supplement other techniques and their own judgment in their attempts to develop realistic and useful cost of equity calculations.

CAPM is based on the following assumptions:

- Investors are risk averse.
- Security returns are normally distributed.
- The utility function of investors is quadratic.
- Investors have homogeneous expectations - they have identical subjective estimates of the means, variances, and covariances among returns.
- Investors can borrow and lend freely at a riskless rate of interest.
- The market is perfect: there are no taxes; there are no transactions costs; securities are completely divisible; the market is competitive.
- The quantity of risky securities in the market is given.

Looking at these assumptions, one may feel that the CAPM is unrealistic. However, the value of a model depends not on the realism of its assumptions, but on the validity of its conclusions. Extensive empirical analysis suggests that there is a lot of merit in the CAPM.

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Security Market Line

According to the CAPM, risk and return are related in a linear fashion:

$$E(R_j) = R_f + \beta_j [E(R_M) - R_f]$$

where

$E(R_j)$ is the expected return on security j,

R_f is the risk-free return,

β_j is the beta of security j,

and

E(Rm) is the expected return on the market portfolio.

As per the above relationship, referred to as the security market line, the required return on a security consists of two components:

Risk-free return : R_f

Risk-premium : $\beta_j [E(R_M) - R_f]$

Note that the risk premium is a product of the level of risk, β_j, and the compensation per unit of risk, $[E(R_M) - R_f]$.

To illustrate, let us consider an example. Stock j has a beta of 1.4. If the risk-free rate is 10 percent and the expected return on the market portfolio is 15 percent, the expected return on stock j is:

10 + 1.4 (15 -10) = 17 percent

It is obvious that, ceteris paribus, the higher the beta, the higher the expected return, and vice versa.

Exhibit Below shows the security market line for the basic data given above. In this figure, the expected return on three securities A, B and C is shown. Security A is a defensive security with a beta of 0.5. Its expected rate of return is 12.5 percent. Security B is a neutral security with a beta of 1. Its expected rate of return is equal to the expected rate of return on the market portfolio namely 15.0 percent. Security C is an aggressive security with a beta of 1.5. Its expected rate of return is 17.5 percent. (In general, if the beta of a security is less than 1 it is characterised as defensive; if it is equal to 1 it is characterised as neutral; and if it is more than 1 it is characterised as aggressive.)

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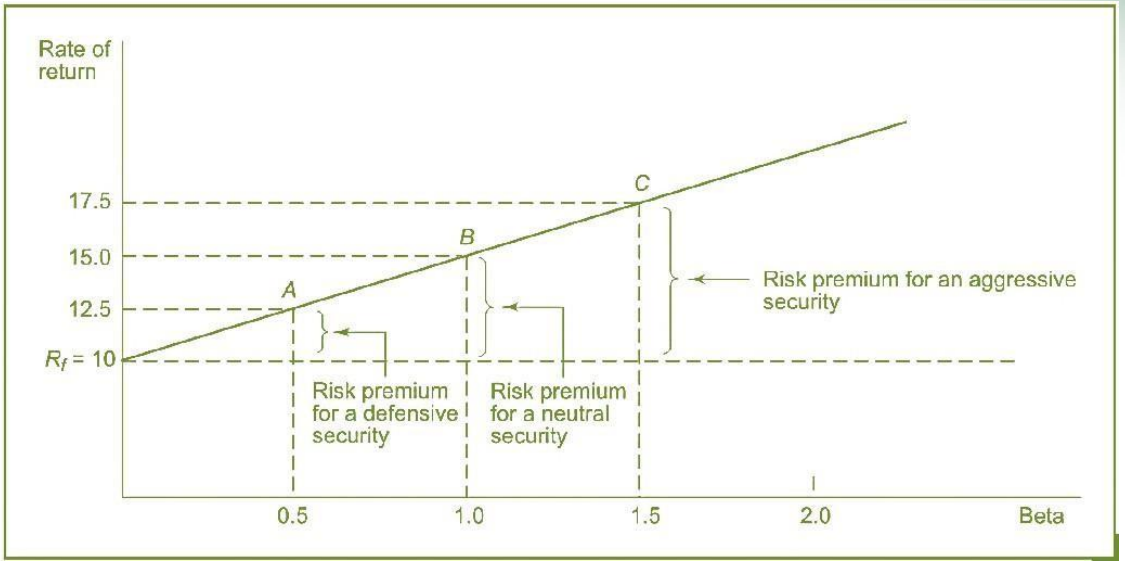
Security Market Line



Changes in Security Market Line

- The two parameters defining the security market line are the intercept (Rf) and the slope [E(Rm)] - R_f]. The intercept represents the nominal rate of return on the risk-free security. It is expected to be equal to the risk-free real rate of return plus the inflation rate.
- For example, if the risk-free real rate of return is 4 percent and the inflation rate is 8 percent, the nominal rate of return on the risk-free security is expected to be 12 percent. The slope represents the price per unit of risk and is a function of the risk-aversion of investors.
- If the real risk-free rate of return and/or the inflation rate changes, the intercept of the security market line changes. If the risk aversion of investors changes, the slope of the security market line changes.
- Exhibit below (A) shows the change in the security market line when the inflation rate increases and
- Exhibit below (B) shows the change in the security market line when the risk-aversion of investors decreases.

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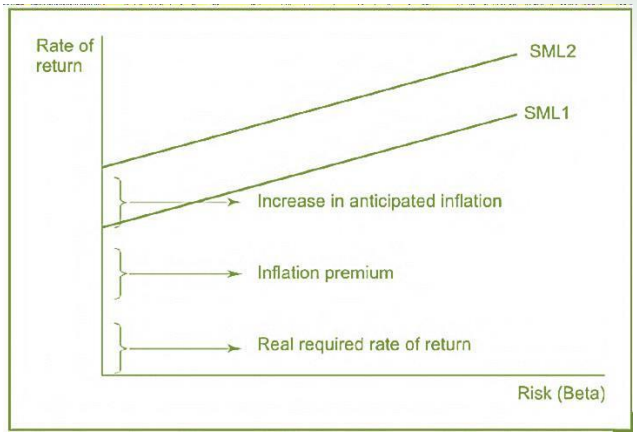
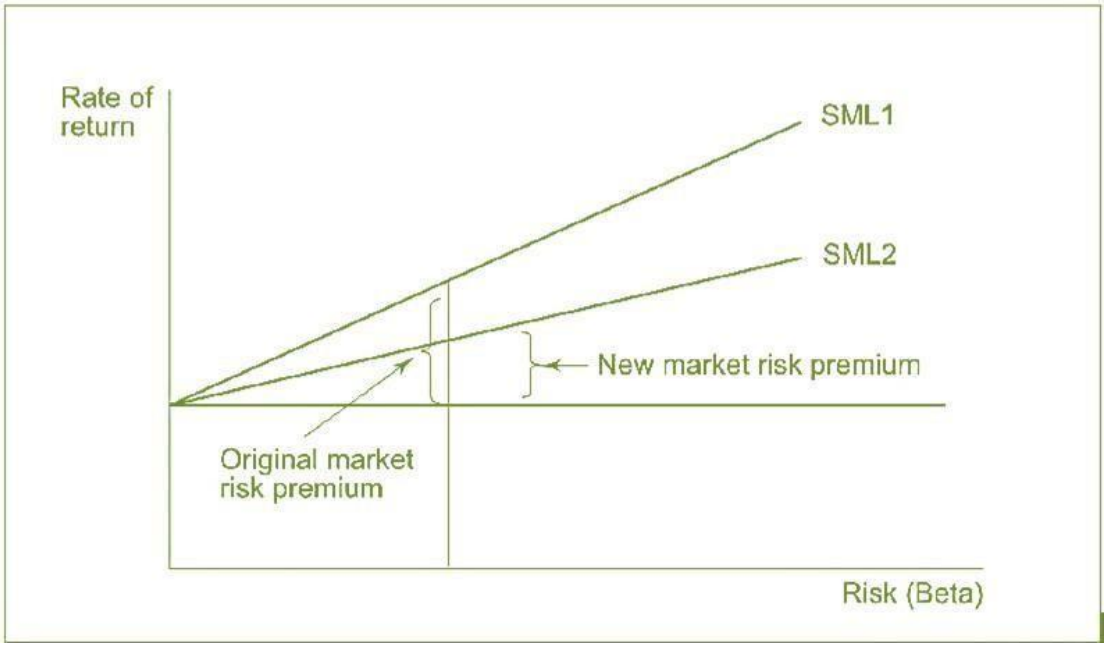
(A) Change in the Security Market Line Caused by an Increase in Inflation



(A) Changes in the Security Market Line Caused by a Decrease in Risk Aversion



Changes in Equilibrium Stock Prices

Stock market prices tend to change in response to changes in the underlying factors. To illustrate, let us assume that stock A, described above, is in equilibrium and sells at a price of Rs.20.00 per share. If the expectation with respect to this stock is fulfilled, its equilibrium price a year hence will be Rs.21.20, six percent higher than the current price. However, several factors could change in the course of a year and alter its equilibrium price. Suppose the values of underlying factors change as follows:

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	Value of the Underlying Factor	
	Original	*Revised*
R_f (Riskless rate)	10%	9%
$R_M - R_f$ (Market risk premium)	4%	3%
β_A (Stock A's beta coefficient)	1.25	1.33
g (Stock A's expected growth rate)	6%	8%

The changes in the first three factors cause R_A to change from 15 percent to 13 percent.

Original : $R_A = 10\% + 1.25\,(4\%) = 15\%$

Revised : $R_A = 9\% + 1.33\,(3\%) = 13\%$

The change in the expected growth rate along with the change in the expected return alters the equilibrium price.

$$\text{Original} : P_o = \frac{1.70(1.06)}{0.15 - 0.06} = \frac{1.80}{0.09} = \text{Rs.}20.00$$

$$\text{Revised} : P_o = \frac{1.70(1.08)}{0.13 - 0.08} = \frac{1.84}{0.05} = \text{Rs.}36.80$$

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9. Assignments (For higher level learning and Evaluation - Examples: Case study, Comprehensive design, etc.,)

S. No.	Questions	K Level	CO Level
1	Compare Preferred Stock Valuation & Common Stock Valuation with Practical examples	K2	CO2
2	Interpret Capital Asset Pricing Model (CAPM) with an Application	K3	CO2

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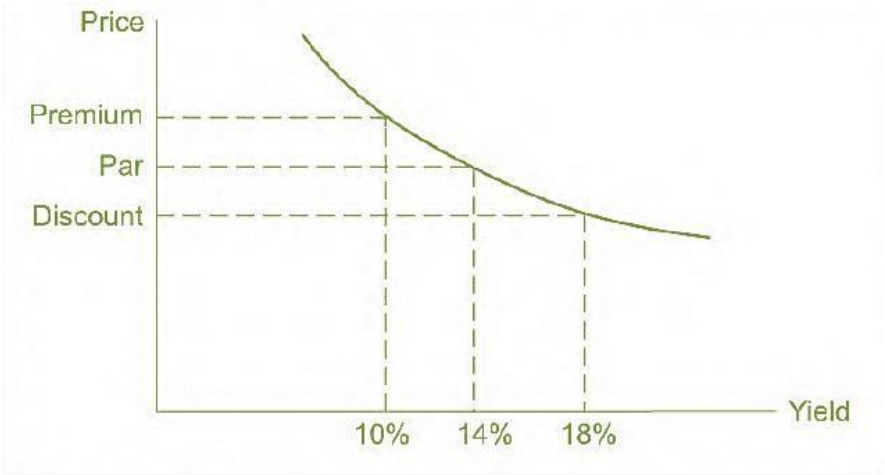
Part A

Sl. No	Questions and Answer	CO	K Level
1	**What is a Bond?** A bond represents a contract under which a borrower promises to pay interest and principal on specific dates to the holders of the bond. Bonds are issued by a variety of organisations. The principal issuers of bonds in India are the central government, state governments, public sector undertakings, private sector companies, and municipal bodies.	CO2	K1
2	**What is par value?** This is the value stated on the face of the bond. It represents the amount the firm borrows and promises to repay at the time of maturity. Usually the par or face value of bonds issued by business firms is Rs.100. Sometimes it is Rs.1,000.	CO2	K1
3	**Compare Public sector bonds and private sector bonds** • **Public sector bonds** bonds are bonds issued by companies in which the central or state governments have an equity stake in excess of 50 percent. Some of these bonds enjoy a tax-free status whereas others are taxable. • **Private sector bonds** are bonds issued by private sector companies. Bonds issued by companies, PSU bonds as well as private sector bonds, generally have maturity ranging from 1 year to 15 years and pay interest semi-annually	CO2	K1
4	**Compare Liquidation Value versus Going Concern Value** • The liquidation value is the amount that can be realised when an asset, or a group of assets representing a part or even the whole of a firm, is sold separately from the operating organisation to which it belongs. • The going concern value represents the amount that can be realised if the firm is sold as a continuing operating entity.	CO2	K1

Part A

Sl. No	Questions and Answer	CO	K Level
5	**Distinguish Liquidation Value versus Going Concern Value** **The liquidation value** is the amount that can be realised when an asset, or a group of assets representing a part or even the whole of a firm, is sold separately from the operating organisation to which it belongs. The **going concern value** represents the amount that can be realised if the firm is sold as a continuing operating entity.	CO2	K1
6	**Compare Book Value versus Market Value** The **book value of an asset** is the accounting value of the asset, which is simply the historical cost of the asset less accumulated depreciation or amortisation as the case may be. The book value of a firm's equity is equal to the book value of its assets minus the book value of its liabilities. The **market value of an asset** is simply the market price at which the asset trades in the market place. Often the market value is greater than the book value.	CO2	K1
7	**Depict the Graph of Price-Yield Relationship** 	CO2	K1

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Sl. No	Questions and Answer	CO	K Level
8	**Define Current Yield** The current yield relates the annual coupon interest to the market price. It is expressed as: $$\text{Current yield} = \frac{\text{Annual interest}}{\text{Price}}$$ The current yield calculation reflects only the coupon interest rate. It does not consider the capital gain (or loss) that an investor will realise if the bond is purchased at a discount (or premium) and held till maturity	CO2	K1
9	**Define Yield to Maturity (YTM)** The yield to maturity (YTM) of a bond is the interest rate that makes the present value of the cash flows receivable from owning the bond equal to the price of the bond. Mathematically, it is the interest rate (r) which satisfies the equation: $$P = \frac{C}{(1+r)} + \frac{C}{(1+r)^2} + \ldots + \frac{C}{(1+r)^n} + \frac{M}{(1+r)^n}$$ where P is the price of the bond, C is the annual interest (in rupees), M is the maturity value (in rupees), n is the number of years left to maturity	CO2	K1
10	**What do you mean by Yield to Call?** • bonds carry a call feature that entitles the issuer to call (buy back) the bond prior to the stated maturity date in accordance with a call schedule (which specifies a call price for each call date).	CO2	K1

Sl. No	Questions and Answer	CO	K Level
11	**What is Preference stock?** Preferred stock generally pays regular, fixed dividends. Preference dividends are not increased when the profits of the firm rise, nor are they lowered or suspended unless the firm faces financial difficulties.	CO2	K1
12	**Define Common stock valuation** Common stock valuation is the process of determining the value of a share of stock in a company. The holder of one share in a company that has one million shares outstanding is actually the owner of one-millionth of the company; the value of that share should represent that percentage of the company's worth.	CO2	K1
13	**Discuss about the Portfolio** Portfolio is the combination of two or more securities or assets. A portfolio is composed of two or more securities. Each portfolio has risk-return characteristics of its own. A portfolio comprising securities that yield a maximum return for given level of risk or minimum risk for given level of return is termed as 'efficient portfolio'.	CO2	K1
14	**What is Expected Rate of Return?** The expected rate of return is the weighted average of all possible returns multiplied by their respective probabilities. $E(R) = \sum_{i=1}^{n} p_i R_i$ Where, E(R) is the expected return, R_i, is the return for the ith possible outcome, p_i, is the probability associated with Rjf n is the number of possible outcomes.	CO2	K1

Sl. No	Part B - Questions	K Level
1	What do you infer from Bond Valuation? Explain in detail.	K2
2	Explain the Risk and Return in a Portfolio Context.	K2
3	Explain the Probability Distributions to Measure Risk.	K2
4	Explain the Expected rate of return in Probability Distributions to Measure Risk.	K2
5	Discuss the relationship between Coupon Rate, Required yield, and Price.	K2
6	Interpret the relationship between Bond Price and Time.	
7	Explain Yield to Maturity with an example and also discuss Yield to call.	K2
8	Discuss the Capital Asset Pricing Model and its assumptions.	K2
9	Compare Public sector bonds and private sector bonds Compare Liquidation Value versus Going Concern Value Compare Book Value versus Market Value	K2
10	Compare Common stock valuation and Preferred Stock valuation	K2

12. Supportive online Certification courses (NPTEL, Swayam, Coursera, Udemy, etc.,)

S.No	Name of the Course	Link
1	Financial Management for Managers	https://nptel.ac.in/courses/110/107/110107144/
2	Finance for Non-Finance Professionals	https://www.coursera.org/learn/finance-for-non-finance
3	Introduction to Finance: The Basics	https://www.coursera.org/learn/introduction-to-finance-the-basics
4	Financial Management	https://www.coursera.org/specializations/financial-management
5	Foundations of Finance	https://www.edx.org/course/foundations-of-finance
6	Finance Essentials	https://www.edx.org/course/finance-essentials
7	Foundations of Finance	https://www.edx.org/course/foundations-of-finance
8	Portfolio Management, Governance, & the PMO	https://www.edx.org/course/portfolio-management-governance-the-pmo
9	Financial Planning for Young Adults	https://www.coursera.org/learn/financial-planning

13. Real time Applications in day to day life and to Industry

Stock Market Quotations and Stock Market Indices

Information on stock market activity is reported in various media. It is covered by on-line services, newspapers, business periodicals, other publications, radio, and television. For most of the investors, the coverage in *The Economic Times* or *The Financial Express* or *Business Standard* is reasonably adequate.

Investors are interested in knowing what is happening to individual stocks and what is happening to the market as a whole. Let us see how the information about these aspects is reported.

Individual Stock Quotations The manner in which daily newspapers provide information on individual stocks may be illustrated with the help of the following quotations of Bajaj Auto from a recent issue of *The Economic Times*. Note that it is a combined share quotation of the BSE and the NSE. The BSE quotes are given first and NSE quotes follow.

The first column gives the name of company. The second column gives the previous closing quotation. The third, fourth, fifth, and the sixth columns give respectively, the opening, high, low, and closing quotations of the day. The seventh column gives the volume of trading, the eighth the number of trades, and the ninth the closing P/E multiple of the day. Finally, the tenth column gives the 52-week high and low.

Company	Prev.cl.	Open	High	Low	Close	Volume	Trades	P/E	52-Wk H/L
Bajaj Auto [BSE]	1577.75	1600	1600	1553.25	1568.50	25K	1465	61.6	1627/700
[NSE]	1574.60	1600	1600	1553.35	1567.80	207K	12574	61.6	1630/723

Stock Market Indices Investors often ask the question: How is the market doing? This interest in the broad market movement stems from the general observation that prices of most of the stocks tend to move together, a fact that has a fairly strong empirical underpinning. The general movement of the market is measured by indices representing the entire market or important segments thereof.

The two most popular stock market indices in India are Sensex and Nifty. A brief discussion of them follows:

- *Bombay Stock Exchange Sensitive Index* Perhaps the most widely followed stock market index in India, the Bombay Stock Exchange Sensitive Index, popularly called the *Sensex* reflects the movement of 30 sensitive shares. The index of any trading day reflects the aggregate market value of the floating stock of the sample of 30 shares on that day in relation to the average aggregate market value of the floating stock of these shares in the base year, 1978-79. The base value of this index is 100.

- *S&P CNX Nifty* Perhaps the most rigorously constructed stock market index in India, the *Nifty* reflects the price movement of 50 shares selected on the basis of market capitalisation and liquidity (impact cost). The index of any trading day reflects the aggregate market value of the floating stock of a sample of 50 shares on that day in relation to the aggregate market value of the floating stock of those shares on November 3, 1995. The base value of this index is 1000.

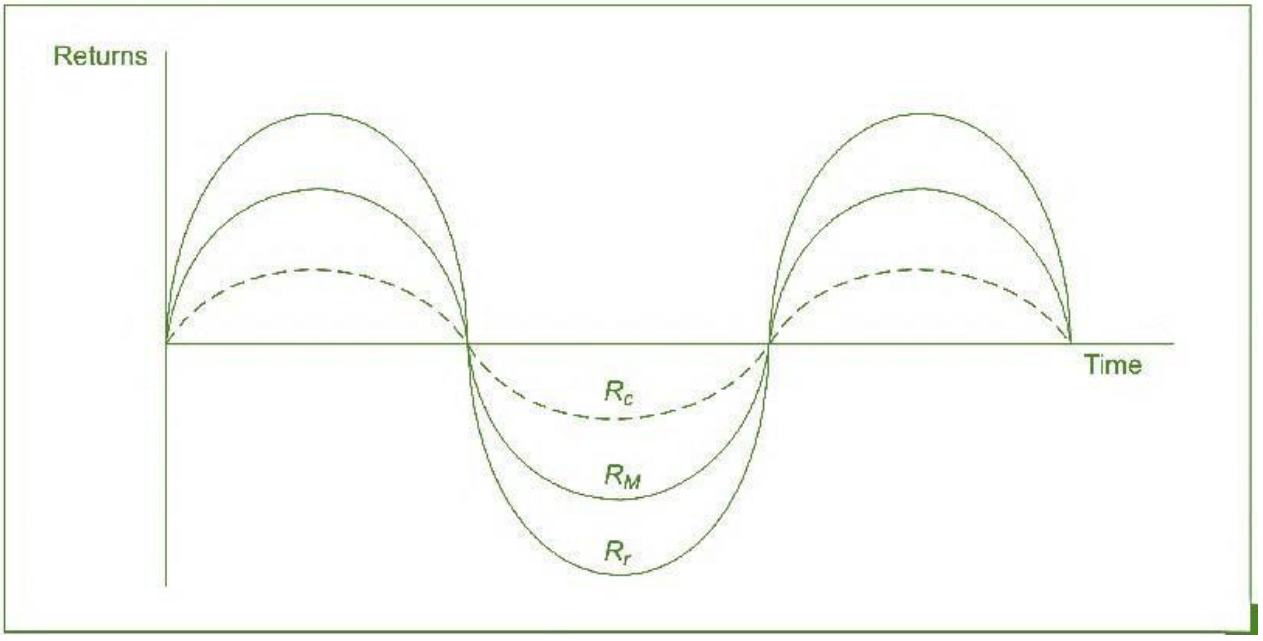
MEASUREMENT OF MARKET RISK

If you want to figure out the contribution of an individual stock to the risk of a well-diversified portfolio, you have to look at its market risk, and not its risk if it is held in isolation.

The market risk of a security reflects its sensitivity to market movements. Different securities seem to display differing sensitivities to market movements. This is illustrated graphically in Exhibit which shows the returns on the market portfolio (R_M) over time, along with returns on two other securities - a risky security, whose return is denoted by (R_r) and a conservative security, whose return is denoted by (R_c). It is evident that R_r is more volatile than R_M whereas R_c is less volatile than R_M.

The sensitivity of a security to market movements is called beta (β). By definition, the beta for the market portfolio is 1. A security which has a beta of, say, 1.5 experiences greater fluctuation than the market portfolio. More precisely, if the return on market portfolio is expected to increase by 10 percent, the return on the security with a beta of 1.5 is expected to increase by 15 percent (1.5 x 10 percent). On the other hand, a security which has a beta of, say, 0.8 fluctuates lesser than the market portfolio. Individual security betas generally fall in the range 0.30 to 2.00 and rarely, if ever, assume a negative value.

Behaviour of Returns Over Time



15. Assessment Schedule

Sl. No	Proposed Date	Actual Date
1	FIAT-	
2	SIAT –	
3	Model –	

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16. PRESCRIBED TEXT BOOKS & REFERENCE BOOKS

TEXT BOOKS:

1. Chandra, Prasanna - Financial Management - Theory & Practice, Tata McGraw Hill, 2007.

2. Srivastava, Misra: Financial Management, OUP, 2011

REFERENCES BOOKS :

1. Van Horne and Wachowicz : Fundamentals of Financial Management, Prentice Hall/ Pearson Education.

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17. MINI PROJECT SUGGESTIONS

- Articulate the Bond and Stock valuation with examples of your choice.
- Interpret risk & return of portfolio, considering the stocks of few business firms

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Thank you

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